

1. Abstract

This study aims to model and examine how the relationship of HIV-1 RNA and CD4 T-cell counts evolve over time at both the individual patient level and across the overall population. The data for this study was drawn from the ATCG Protocol 315 which provides longitudinal measurements of HIV-1 RNA and CD4 counts for infected patients across multiple time points. Assessing the dynamic interaction between HIV-1 RNA and CD4 is essential for understanding disease progression following antiviral treatment.

Using longitudinal statistical methods, the relationship between HIV-1 RNA and CD4 was analyzed through both Linear Mixed Effect models (LMM) and Nonlinear Mixed Effect models (NLME). Model performance was evaluated using diagnostic measures as well as Akaike Information Criterion (AIC). The results indicate nonlinear models provided a better fit to the data with more stable diagnostics and an improved ability to capture the underlying nonlinear relationship between RNA and CD4 that linear models failed to capture. Among the four models considered, Model D was selected as the best overall based on better fitted diagnostics and lower AIC values. These findings highlight the importance of flexible modeling approaches in capturing complex biological relationships and contribute to understanding of HIV-1 progression.

2. Introduction

Human Immunodeficiency Virus (HIV) infection is characterized by changes in viral load and immune system function over time. HIV-1 RNA levels provide a measure of viral load, while CD4 T-cell counts are an important indicator of immune response. Understanding how these measures evolve over time is critical for evaluating disease progression and treatment effectiveness.

Longitudinal data, where repeated measurements are collected from the same individuals over time, require methods that account for within-subject correlation and variability. In particular, individual patients may exhibit different response patterns, making it important to incorporate subject-specific effects into the analysis.

The objective of this analysis is to investigate the relationship between HIV-1 RNA, time (in days), and CD4 T-cell counts using both linear and nonlinear mixed-effects models to capture potential nonlinear trends and variability across individuals.

3. Data Sources

The dataset used for the study was drawn from ACTG Protocol 315, a clinical study designed to evaluate the effects of antiviral regimen in HIV-infected patients. The dataset consists of longitudinal measurements of HIV-1 RNA and CD4 T-cell counts collected following the treatment. Measurements were recorded at days 0, 2, 7, 10, 14, 21, and 28, as well as at weeks 8, 12, 24, and 48.

The original study consisted of 53 infected patients who were administered the antiviral regimen. However, five patients discontinued the study and two patients had missing data, resulting in final sample of 46 patients for analysis. Data beyond 3 months were excluded as they were likely to be contaminated due to long-term clinical factors.

Key variables in the dataset include, HIV-1 RNA, which measures viral load (measured on $\log_{10}$ scale) and CD4 T-cell count, which serves as indicator for assessing immunological response. Additional

variables include patient identifiers and time of measurement (recorded in days) to provide tracking of individual measurements over time.

4. Methodology

Preprocessing:

Prior to exploratory data analysis, preprocessing steps were performed to ensure data quality and suitability for analysis. The dataset was subset to include only observations within the three-month (≤ 91 days) study period. Following subsetting, variables were encoded as numerical or categorical features where appropriate. This ensures that, during model fitting, the longitudinal modeling procedures (LMM and NLME) correctly associate repeated measurements with the corresponding patient.

Exploratory Data Analysis:

1. Summary Statistics:

Following preprocessing, summary statistics were generated to examine the distribution, central tendency, and variability of the data. These statistics were computed at both the individual patient level and for the overall population, and included count, mean, standard deviation, quantiles, minimum, and maximum values. These metrics are provided useful as they provide insight into exploratory analysis and identify irregularities in the data.

2. Spaghetti Plots:

Spaghetti plots were used to assess relationships among HIV-1 RNA, CD4, and time prior to model fitting. These plots visualize the relationship between two variables by mapping observations along the x- and y-axes. A LOESS smoother was added to highlight underlying trends and identify the appropriate functional form (e.g., linear vs. nonlinear).

Additionally, spaghetti plots were generated to visualize both individual patient trajectories and overall population trends over time. Overlaying a LOESS curve on these plots allowed for comparison between individual-level patterns and the population trend, helping to assess whether patients follow a common trajectory or exhibit deviation from the overall pattern.

3. Violin Plots:

To examine the distribution and shape of the data, violin plots were utilized. These plots combine features of density plots and boxplots, providing insight into both the distribution and spread of the data. Overlaid boxplots were included to highlight measures of central tendency, variability, and potential outliers. Violin plots were used to examine the distribution of CD4 and HIV-1 RNA across time to identify symmetry or skewness of the distribution as well as potential outliers.

4. Correlation Matrices/Correlation Plots:

Correlation matrices and corresponding visualizations were used to examine relationships among continuous variables. High correlations between covariates may indicate redundancy, where variables capture similar information. This can lead to instability in coefficient estimates, including inflated standard errors and unreliable inference.

Candidate Models:

1. Model A: Linear Mixed Effect (Random Intercept)

Model A was specified as a Linear Mixed Effect model with a random intercept:

$$V_{ij} = \beta_1 + \beta_2 t_{ij} + \beta_3 t_{ij} CD4_{ij} + \mu_i + e_{ij}$$

This model was considered a candidate model because it accounts for patient-specific differences in baseline HIV-1 RNA measurements. Exploratory spaghetti plots indicated patients exhibited variability in their initial HIV-1 RNA and CD4 values suggesting inconsistent baseline levels across patients. The random intercept allows each patient's baselines to shift above or below the population average while the effect of time for each patient remains constant across patients. Thus, Model A allows for the variation in baseline levels while assuming a common slope over time providing a candidate model for the data.

2. Model B: Linear Mixed Effect (Random Coefficients)

Model B was specified as a Linear Mixed Effect model with random coefficients:

$$V_{ij} = \beta_{i1} + \beta_{i2} t_{ij} + \beta_{i3} t_{ij} CD4_{ij} + e_{ij}$$

where the subject-specific coefficients are defined as:

$$\beta_{ij} = \beta_j + \mu_{ij}, \quad j = 1, 2, 3$$

Model B was considered a candidate model because it accounts for patient-specific differences in both baseline HIV-1 RNA measurements and the trajectory of measurements over time. Exploratory spaghetti plots indicated patients exhibited variability in both their initial HIV-1 RNA and CD4 values and their measurements across multiple timepoints. While many patients followed the overall population trend initially, over time deviation was observed suggesting variability in individual slopes. The inclusion of the random coefficient allows the effect of time to differ for each patient meaning each patient's slope is allowed to vary over the 91-day period. As an extension of Model A, Model B's inclusion of random coefficients allows for variability in both baseline level and trajectories over time.

3. Model C: Nonlinear Mixed Effect (Random Intercept)

Model C was specified as a Nonlinear Mixed Effect model with a random intercept:

$$V_{ij} = \beta_1 e^{\{-\beta_2 t_{ij}\}} + \beta_3 e^{\{-\beta_4 CD4_{ij}\}} + \mu_i + e_{ij}$$

Given the nonlinear curves exhibited from the exploratory scatterplots, nonlinear mixed effect models were considered to better capture the relationship among HIV-1 RNA, time, and CD4. Model C was specified by incorporating exponential terms while allowing patient-specific baseline differences through a random intercept. Similar to Model A, random intercepts were fitted first to account for variability in patients' initial HIV-1 RNA and CD4 measurements before extending the nonlinear structure to include random coefficients. The exponential form was used to capture the nonlinear relationship between HIV-1 RNA and CD4 suggested by the exploratory analysis.

4. Model D: Nonlinear Mixed Effect (Random Coefficient)

Model D was specified as a Nonlinear Mixed Effect model with random coefficients:

$$V_{ij} = \beta_1 e^{\{-\beta_2 t_{ij}\}} + \beta_3 e^{\{-\beta_4 CD4_{ij}\}} + e_{ij}$$

where the subject-specific coefficients are defined as:

$$\beta_{ij} = \beta_j + \mu_{ij}, \quad j = 1, 2, 3, 4$$

Model D extends Model C by allowing model parameters to vary across patients, thereby capturing both nonlinear relationships and individual-specific trajectories. While Model C accounts for variability in baseline levels through a random intercept, Model D further incorporates random coefficients to allow patient-specific variation in both the baseline measurements and slopes over time.

Consistent with the patterns observed in exploratory spaghetti plots, patients exhibited variability not only in their initial HIV-1 RNA and CD4 values but also in how these measurements evolved over time. The inclusion of random coefficients allows the nonlinear trajectories to differ across individuals, capturing deviations from the overall population trend.

The exponential functional form was retained from Model C to capture the nonlinear relationships between HIV-1 RNA, time, and CD4, reflecting the complex nature of the underlying biological processes.

Model Diagnostic Plots:

Model assumptions were assessed using diagnostic plots for both linear mixed-effects (LMM) and nonlinear mixed-effects (NLME) models. The primary diagnostics included residuals versus fitted values plots and Q-Q plots.

1. Linear Mixed Effect models

For the linear mixed-effects models, residuals versus fitted values plots were used to assess the assumption of linearity and to evaluate homoscedasticity. Ideally, these plots should display residuals randomly scattered around zero with no obvious pattern. Systematic patterns, such as curvature, indicate a violation of the linearity assumption and suggest model misspecification. Additionally, non-constant spread in the residuals may indicate heteroscedasticity.

Q-Q plots were used to assess the normality of both residuals and random effects. Points lying approximately along the reference line indicate that the normality assumption is reasonably satisfied, while substantial deviations suggest departures from normality.

2. Nonlinear Mixed Effect models

For the nonlinear mixed-effects models, residuals versus fitted values plots were used to evaluate the suitability of the specified functional form. As with the linear models, residuals should be randomly scattered around zero without systematic patterns. The exhibition of structured patterns would indicate that the nonlinear function may not adequately capture the relationship in the data.

Q-Q plots were again used to assess the normality of residuals and random effects. Approximate alignment of the points along the reference line suggests that the normality assumption is reasonable.

Model Comparison:

Across the four models fitted, Akaike Information Criterion (AIC) was used for model comparison. For this comparison, all models were fit using maximum likelihood (ML), as the linear and nonlinear models differed in their fixed effects. Specifically, the linear models included time and an interaction term between time and CD4, whereas the nonlinear models incorporated only time and CD4 differing in fixed effects. Thus, the model with the lowest AIC value was considered to provide the best fit to the data.

5. Analysis

All analyses of this study were conducted using R. Data manipulation and preprocessing were performed using the *tidyverse* suite of packages, including **dplyr** and **tidyr**, which were used for data cleaning, transformation, and reshaping. Data visualization was carried out using **ggplot2**, with additional support from **ggsci** for enhanced graphical aesthetics and **corrplot** for correlation analysis. Longitudinal statistical modeling was performed using both linear and nonlinear mixed-effects approaches, implemented through the **lme4** and **nlme** packages. Model comparison and selection were conducted using Akaike Information Criterion (AIC) computed using base R functions.

Category	Package/Libraries	Purpose/Implementation:
Data Manipulation	tidyverse, tidyr, dplyr	Utilized for preprocessing data for visuals and modeling.
Statistical Summary	stats	Implemented to generate descriptive summary statistics.
Visualization	ggplot2, ggsci, corrplot	Create visualizations to examine relationships between variables.
Statistical Models	lme4, nlme	Applied to fit linear and nonlinear models

Table 1: R Libraries and Packages utilized for analysis

Preprocessing:

The dataset was subset to include only observations from the period of 0-91 days. Since, time was recorded in both days and weeks, days were used as the primary unit of measurement to allow for greater precision. Using weeks as a unit of measure of time would provide up until 84 days. While a 12-week cutoff corresponds to 84 days, this would have excluded some relevant observations. Therefore, a cutoff of ≤ 91 days was selected, which allowed inclusion of all valid measurements within the early treatment window, including a key observation for patient 11542 at day 91.

Following subsetting, the identifier variable (**patid**) was re-encoded as a categorical variable. This ensures during the modeling process both longitudinal modeling procedures (**lme** and **nlme**) correctly associate repeated measurements with the corresponding patient. Additionally, two indexing variables present in the original dataset were not used for the analysis and thus removed from the subset.

Exploratory Data Analysis:

1. Summary Statistics

Variable	n	mean	sd	min	p25	median	p75	max	NA
Day	329	24.410334	27.327932	0.00000	6.000000	12.000000	29.000000	91.00000	0
cd4	329	245.902097	95.703878	17.28000	183.150000	245.700000	314.160000	488.40000	0
Log ₁₀ RNA	329	3.676374	1.106829	1.69897	2.857332	3.662758	4.544068	6.20412	0

Table 2: Descriptive summary statistics for variables Day, cd4, and HIV-1 RNA.

Table 2 summarizes the distribution of key variables used in the analysis, based on a total of $n = 329$ observations. The time variable, Day, ranges from 0-91, with a mean of 24.4 and a median of 12 indicating a majority of the observations occur earlier in the study.

The distribution of CD4 appears approximately symmetric as both mean (245.9) and median (245.7) are similar in value. The values range from 17.3 to 488.4 with a standard deviation of 95.7, suggesting moderate variability in immune response.

HIV-1 RNA is measured on the log₁₀ scale. The distribution of log₁₀ scale also appears approximately symmetric, with a mean of 3.68 and a median of 3.66. Values range from 1.7 to 6.2, corresponding to 50 to 1,600,000 on the original scale. While HIV-1 RNA appears to have moderate variability ($sd = 1.11$) on the log-transformed scale, the original scale shows a substantial variability in viral load.

2. Spaghetti Plots

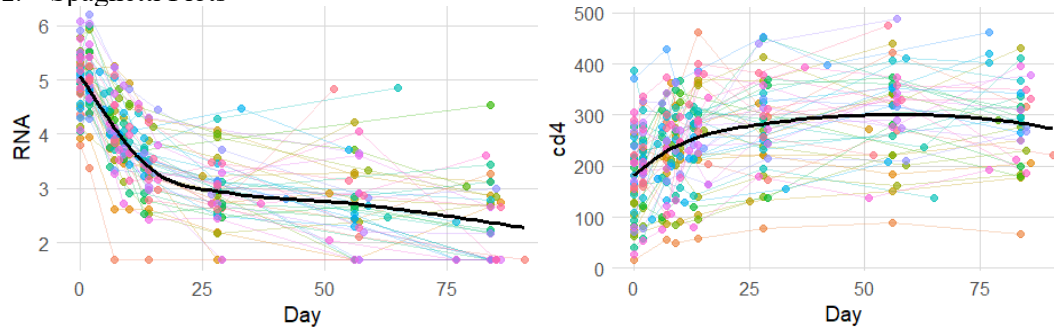


Figure 1: Comparison of individual trajectories for HIV-1 RNA and CD4 over the 91 day period to the overall population trend.

From Figure 1, the spaghetti plots provide further insight into the longitudinal behavior of HIV-1 RNA and CD4 across patients. Both plots exhibit nonlinear trends, suggesting that linear models may be insufficient and that nonlinear models are more appropriate for capturing the underlying relationships. While many patients initially follow the overall population trend, increasing deviations over time indicate variability in individual trajectories, supporting the use of random coefficients in subsequent modeling.

The plots also suggest an inverse relationship between HIV-1 RNA and CD4. Following initiation of antiviral treatment, RNA levels decrease rapidly in the early days and then gradually decline over time, indicating a stronger treatment effect in the initial phase and a diminishing effect after. In contrast, CD4 counts increase during the early period of the study and approach a plateau, with a slight decline observed in the later stages, suggesting stabilization of the treatment effect.

3. Violin Plots

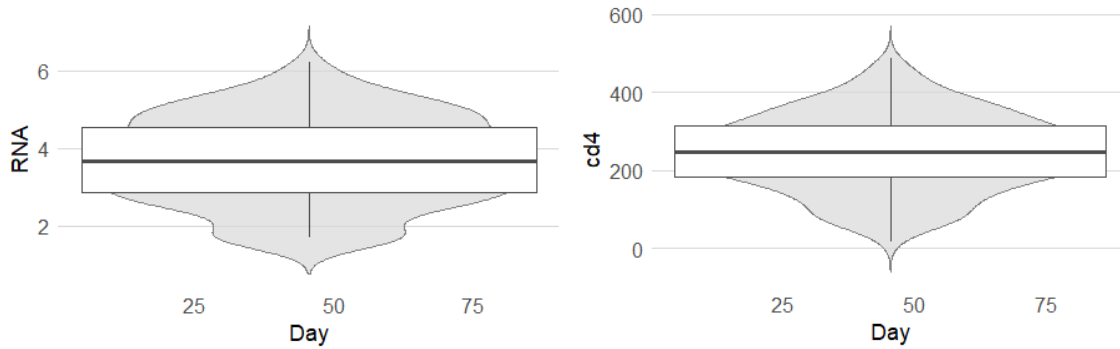


Figure 2: Violin plots overlaid with boxplots illustrating the distribution and variability of HIV-1 RNA ($\log_{10}$ scale) and CD4 counts over time.

Figure 2 illustrates the distribution and shape of both HIV-1 RNA and CD4 counts across time.

Both variables exhibit approximately symmetric distributions which is consistent with earlier findings from the summary statistics where both mean and medians were similar. However, HIV-1 RNA shows a slightly greater spread compared to CD4, indicating more variability in viral load over time. There is also mild evidence of right skewness in both variables, as suggested by longer upper whiskers in the boxplots.

Overall, these visualizations support the numerical summaries and highlight differences in variability between the two measures. While no extreme outliers are evident beyond the whiskers, the distributions suggest that RNA values exhibit a wider spread than CD4 measurements.

4. Correlation Matrices/Correlation Plot

To assess the relationships between HIV-1 RNA, time (in days), and CD4 counts, Pearson correlation coefficients were computed. Correlations range from -1 to 1, where values closer to -1 indicate a strong negative relationship and values closer to 1 indicate a strong positive relationship.

The correlation matrix (Figure A1) summarizes the pairwise relationships. RNA and Day exhibit a strong negative relationship ($\rho = -0.70$) indicating levels tend to decrease over time. This finding is consistent with trends observed in the spaghetti plots.

CD4 and Day show a moderate positive correlation ($\rho = 0.33$), suggesting that CD4 counts tend to increase over time. This relationship is also consistent with the overall trends observed in the longitudinal plots.

Additionally, HIV-1 RNA and CD4 exhibit a moderate negative correlation ($\rho = -0.42$), reflecting the expected inverse relationship between viral load and immune response.

Model Fitting & Diagnostic Plots:

1. Model A

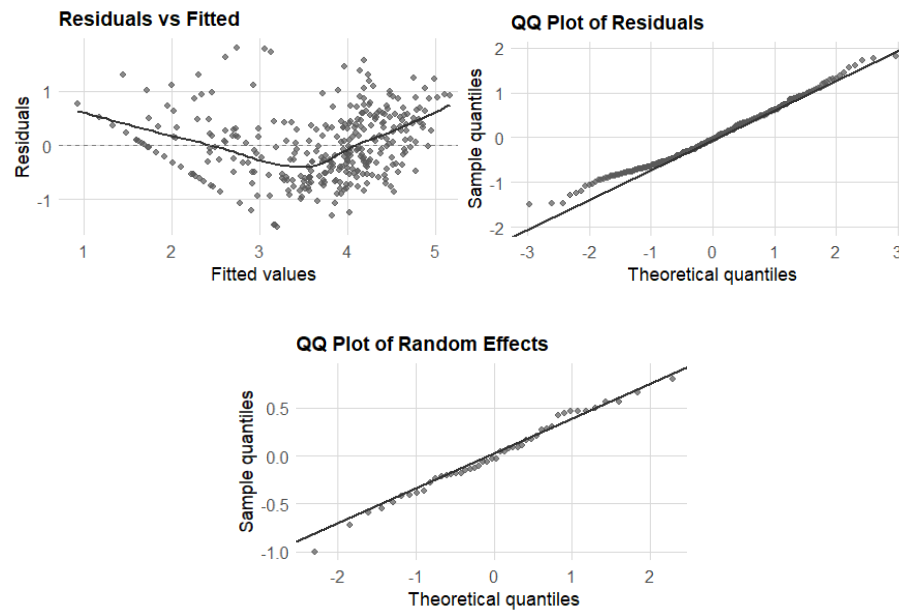


Figure 3: Diagnostics for Model A including Residual vs Fitted and Q-Q plots.

Diagnostic plots shown in Figure 3 are used to evaluate how well Model A fits the data.

The residuals versus fitted plot exhibits a clear nonlinear pattern, as evidenced by the U-shaped trend. This suggests that the current functional form of the model may be misspecified and does not fully capture the relationship between HIV-1 RNA and CD4 over time.

The normality assumptions for the residuals and random effects appear to be reasonably satisfied. While slight deviations from the reference line are present in the Q-Q plots, these deviations are not substantial. Most observations lie close to the reference line, indicating only minor departures from normality. Overall, the distributions can be considered approximately normal.

2. Model B

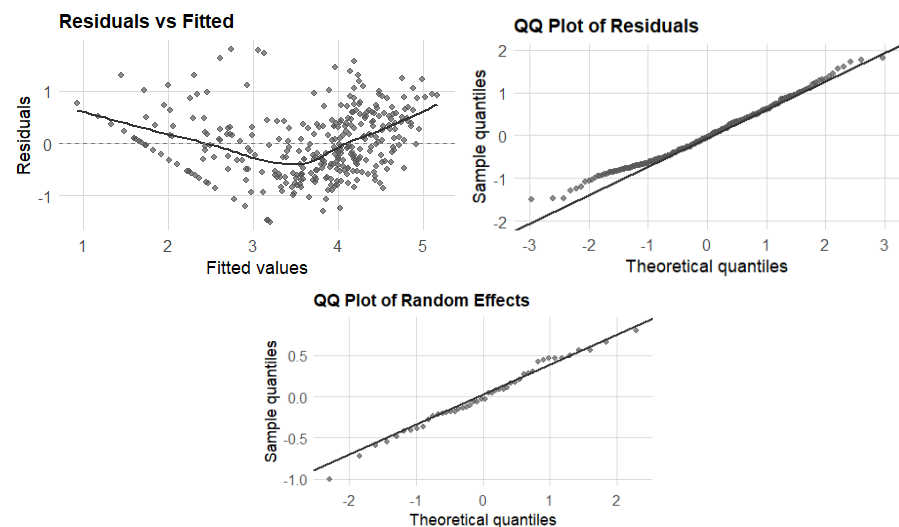


Figure 4: Diagnostics for Model B including Residual vs Fitted and Q-Q plots

Diagnostic plots in Figure 4 indicate modest improvement in model fit compared to Model A.

The residuals versus fitted plot continues to display a clear nonlinear pattern, as evidenced by the U-shaped trend. However, the curvature appears slightly reduced and the tails are more flattened relative to Model A suggesting some improvement in fit. The extension of Model A using random coefficients suggests that the current functional form of the model still may be misspecified and does not fully capture the underlying relationship between HIV-1 RNA and CD4 over time.

The normality assumptions for the residuals and random effects appear similar to those observed in Model A. While minor deviations from the reference line are also present in the Q-Q plots, these deviations are not substantial. Most observations lie close to the reference line indicating the normality assumption is reasonably satisfied.

3. Model C

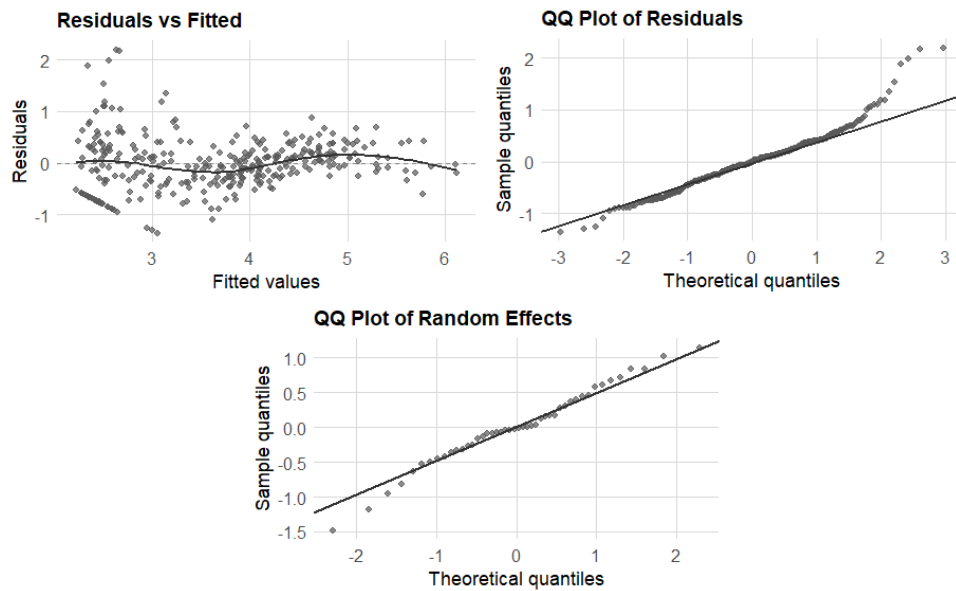


Figure 5: Diagnostics for Model C including Residual vs Fitted and Q-Q plots

Diagnostic plots in Figure 5 illustrate substantial improvement in model fit compared earlier models.

The inclusion of exponential functions in Model C more effectively captures the underlying relationship between HIV-1 RNA and CD4 over time. This is reflected in the residuals versus fitted plot, where the pattern appears largely random with only minor curvature.

The normality assumptions for the residuals and random effects appear to be reasonably satisfied in Model C. Although slight deviations in the tails from the reference line are visible in the Q-Q plots, most observations lie close to the reference line, indicating the normality assumption is not severely violated.

4. Model D

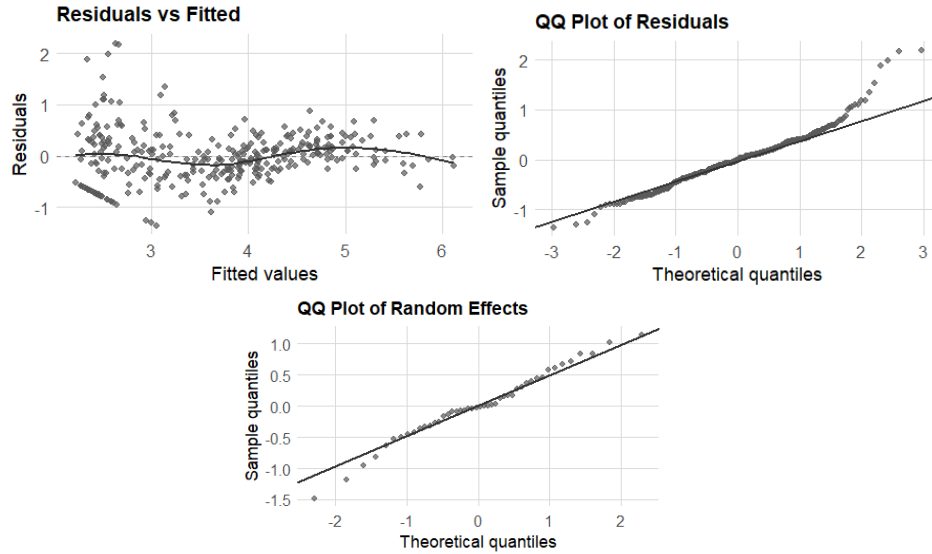


Figure 6: Diagnostics for Model D including Residual vs Fitted and Q-Q plots

Diagnostic plots shown in Figure 6 indicate that Model D provides an improved fit compared to Models A and B, and exhibits similar diagnostic characteristics to Model C.

As an extension of Model C with the inclusion of random coefficients, Model D more effectively captures the underlying relationship between HIV-1 RNA and CD4 over time as well. This is evident in the residuals versus fitted which displays a largely random pattern with only slight curvature.

The Q-Q plots suggest that the normality assumptions for both the residuals and random effects are reasonably satisfied. As with Model C, minor deviations are observed in the tails; however, most observations lie close to the reference line, indicating that these deviations are not substantial.

Model Comparison:

Model:	Converged:	DF:	AIC:	
Model A	Yes	5	730.3239	Linear Mixed Effect (Random Intercept)
Model B	Yes	10	729.8363	Linear Mixed Effect (Random Coefficient)
Model C	Yes	6	605.1156	Nonlinear Mixed Effect (Random Intercept)
Model D	Yes	9	465.6928	Nonlinear Mixed Effect (Random Coefficient)

Table 3: Model Comparison across all models using AIC

Table 3 summarizes the model comparison across all candidate models using Akaike Information Criterion (AIC). Lower AIC values indicate better model fit while accounting for model complexity.

All models successfully converged. Model D yielded the lowest AIC value (465.69), indicating the best overall fit among the models considered. Model C, while showing similar diagnostic behavior, had a substantially higher AIC (605.12), suggesting that incorporating nonlinear effects alone was insufficient to fully capture the variability in the data.

The improvement from Model C to Model D highlights the importance of accounting for subject-specific variability through random coefficients. Similarly, Models A and B, which assume linear relationships, exhibited much higher AIC values, indicating poorer fit relative to the nonlinear models.

The substantial reduction in AIC from Model C to Model D indicates a meaningful improvement in model fit; thus, Model D was selected as the final model.

6. Results

Based on model comparison using AIC and diagnostic assessment, Model D was selected as the final model.

Parameter:	Estimate:	Std. Error:	DF:	t-value	p-value
β_1	2.4362701	0.13672575	280	17.81866	<0.0001
β_2	0.0493600	0.00483409	280	10.21081	<0.0001
β_3	2.7620321	0.16061023	280	17.19711	<0.0001
β_4	0.0005213	0.00018266	280	2.85408	0.0046

Table 4: Parameter estimates for Model D.

Parameter estimates from Model D are presented in Table 4. All estimated parameters are statistically significant ($p < 0.01$), indicating that each term contributes meaningfully to the model. The larger coefficients (β_1 and β_3) primarily influence the overall level of HIV-1 RNA, while the smaller coefficients (β_2 and β_4) reflect more gradual changes over time and with respect to CD4.

The relatively small values of β_2 and β_4 suggest that RNA changes slowly as time progresses and CD4 levels vary. In contrast, β_1 and β_3 contribute more to the overall magnitude of RNA. The small standard errors relative to the estimates indicate that the parameter estimates are stable. Overall, Model D provides a good representation of the relationship between HIV-1 RNA, CD4, and time.

7. Conclusion

This analysis investigated the relationship between HIV-1 RNA, time (in days), and CD4 T-cell counts using longitudinal modeling techniques. Initial exploratory analysis indicated nonlinear patterns and subject-specific variability.

Subsequent modeling and diagnostic assessment identified Model D, a nonlinear mixed-effects model with random coefficients, as providing the best fit to the data. This model most effectively captured the underlying relationships among the variables, as supported by improved diagnostics and the lowest AIC value.

Overall, the results suggest that HIV-1 RNA decreases over time while CD4 counts increase. However, individual trajectories vary, indicating that not all patients follow the same pattern and highlighting variability in subject-specific responses. These findings emphasize the importance of accounting for both nonlinear dynamics and subject-specific variability when modeling complex biological processes.

References

- Chen, J. (2010). *Modelling long-term human immunodeficiency virus dynamic models with application to acquired immune deficiency syndrome clinical study*. Journal of the Royal Statistical Society: Series C (Applied Statistics), 59(5), 805–820.
- Wu, H. (2005). Statistical methods for HIV dynamic studies in AIDS clinical trials. *Statistical Methods in Medical Research*, 14(2), 171–192.
- Perelson, A. S., Neumann, A. U., Markowitz, M., Leonard, J. M., & Ho, D. D. (1996). HIV-1 dynamics in vivo: Virion clearance rate, infected cell life-span, and viral generation time. *Science*, 271(5255), 1582–1586.

Appendix A:

Appendix A provides supplementary exploratory analyses used to investigate the relationships among HIV-1 RNA, time (in days), and CD4 counts.

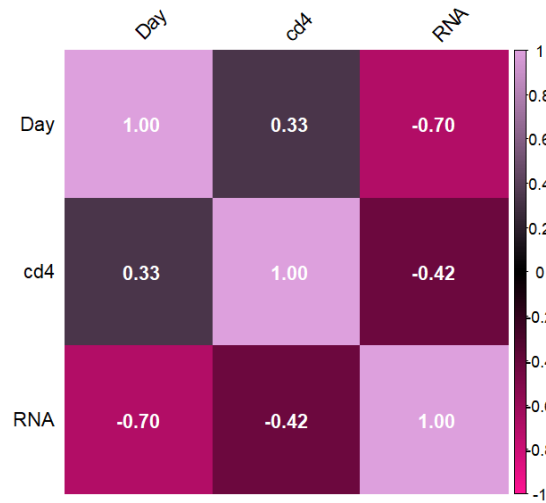


Figure A1: Correlation plot illustrating the strength of relationships among HIV-1 RNA, CD4, and time (in days).

Appendix B:

Appendix B provides R output for Models A, B, and C, including parameter estimates and model summaries.

Parameter:	Estimate:	Std. Error:	DF:	t-value	p-value
Intercept	4.365687	0.08128774	281	53.70658	<0.0001
Day	-0.019893	0.00406838	281	4.88968	<0.0001
Day_cd4	-0.000028	0.00001306	281	-2.11617	0.0352

Table B1: Parameter estimates for Model A.

Parameter:	Variance:	Std. Error:
Intercept	0.1974050	0.4443028
Residual	0.4309824	0.6564925

Table B2: Random Effect estimates for Model A.

Parameter:	Estimate:	Std. Error:	DF:	t-value	p-value
Intercept	4.379648	0.08720097	281	50.22477	<0.0001
Day	-0.024787	0.00617324	281	-4.01519	<0.0001
Day_cd4	-0.000015	0.00001867	281	-0.78051	0.4357

Table B3: Parameter estimates for Model B.

Parameter:	Variance:	Std. Error:
Intercept	2.529875e-01	0.5029785964
Day	6.976458e-04	0.0264129856
Day_cd4	5.709359e-09	0.0000755603
Residual	3.679485e-01	0.6065875994

Table B4: Random Effect estimates for Model B.

Parameter:	Estimate:	Std. Error:	DF:	t-value	p-value
β_1	2.4363566	0.13671266	280	17.82100	<0.0001
β_2	0.0493996	0.00483579	280	10.21542	<0.0001
β_3	2.7620867	0.16057087	280	17.20167	<0.0001
β_4	0.0005208	0.00018261	280	2.85218	0.0047

Table B5: Parameter estimates for Model C.

Parameter:	Variance:	Std. Error:
Intercept	0.3801689	0.6165784
Residual	0.2913355	0.5397550

Table B2: Random Effect estimates for Model C.

Appendix C:

Appendix C provides the R code used for preprocessing, exploratory data analysis, model fitting, diagnostics, and model comparison.

Libraries:

```
packages =  
c("dplyr", "ggplot2", "lme4", "nlme", "tidyverse", "tidyr", "knitr", "ggsci", "corrplot", "stats", "car")  
lapply(packages, require, character.only = TRUE)
```

Preprocessing:

```
#read in data  
aids_data <- read.table("ACTG315.dat", header = TRUE)  
  
#subset the data  
aids_sub <- aids_data %>%  
  filter( Day <= 91) %>%  
  mutate(patid = as.factor(patid)) %>%  
  select(-c("X.No", "No"))
```

Exploratory Data Analysis:

Summary Statistics:

```
summary_stats <- aids_sub %>%  
  summarise( across(c(Day, cd4, RNA),  
    list(  
      n = ~sum(!is.na(.x)),  
      mean = ~mean(.x, na.rm = TRUE),  
      sd = ~sd(.x, na.rm = TRUE),  
      min = ~min(.x, na.rm = TRUE),  
      p25 = ~quantile(.x, 0.25, na.rm = TRUE),  
      median = ~quantile(.x, 0.5, na.rm = TRUE),  
      p75 = ~quantile(.x, 0.75, na.rm = TRUE),  
      max = ~max(.x, na.rm = TRUE),  
      NAcoun = ~sum(is.na(.x)) )  
    )  
  ) %>%  
  pivot_longer(everything(),  
    names_to = c("Variable", ".value"),  
    names_sep = "_" )  
  
summary_stats %>% kable(caption = "Summary Statistics")
```

Spaghetti Plots:

```
# Trajectory trends by all patients over time
scatterplot_patid <- function(df, x, y, z){
  ggplot(df, aes(x = {{ x }}, y = {{ y }}, group = {{ z }}, color =
as.factor({{ z }}})) +
  geom_line(alpha = 0.30, linewidth = 0.4) +
  geom_point(alpha = 0.65, size = 1.8) +
  geom_smooth( aes(group = 1), method = "loess", color = "black",
linewidth = 1, se = FALSE ) +
  theme_minimal(base_size = 12) +
  theme(
    panel.grid.minor = element_blank(),
    panel.grid.major = element_line(color = "grey85", linewidth
= 0.3),
    axis.title = element_text(size = 13),
    axis.text = element_text(size = 11),
    plot.title = element_text(size = 14, face = "bold"),
    legend.position = "none" )
}
scatterplot_patid(aids_sub, Day, RNA, patid)
scatterplot_patid(aids_sub, Day, cd4, patid)
```

Violin Plot:

```
violin_plots <- function(df, x, y){

  ggplot(df, aes(x = {{x}}, y = {{y}})) +
  geom_violin(
    fill = "grey85",
    color = "grey50",
    alpha = 0.7,
    trim = FALSE) +
  geom_boxplot(
    width = 0.1,
    fill = "white",
    color = "grey30",
    outlier.color = "grey40", outlier.size = 1.5 ) +
  theme_minimal(base_size = 12) +
  theme(
    panel.grid.minor = element_blank(),
    panel.grid.major.x = element_blank(),
    panel.grid.major.y = element_line(color = "grey85"),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 11) )
}

violin_plots(aids_sub, Day, RNA)
violin_plots(aids_sub, Day, cd4)
```

Correlation:

```
correlation_matrix <- function(df,x,y,z){  
  cor_mat <- df %>%  
    select({{x}},{{y}},{{z}}) %>%  
    cor(use = "complete.obs")  
  
  cor_plot <- corrplot(cor_mat,  
    method = "color",  
    col = colorRampPalette(c("deeppink", "black", "plum"))(200),  
    addCoef.col = "white",  
    tl.col = "black",  
    tl.srt = 45)  
  
  return(cor_mat)  
  return(cor_plot)  
}  
  
correlation_matrix(aids_sub,Day,cd4,RNA)
```

Model Building:

Functions to call following model fit to asses diagnostics and estimates.

Summary Results

```
summary_res <- function(model){  
  #overall summary outputs  
  summary_model <- summary(model)  
  
  #fixed effects: beta estimates  
  fix_eff <- fixed.effects(model)  
  
  #random effects: BLUPs of mu_i  
  rand_eff <- random.effects(model)  
  
  # var-cov matrix  
  VarCovMat <- vcov(model)  
  VarCorrMat <- VarCorr(model)  
  
  #subject-specific fitted values: Vhat_ij  
  Vij <- fitted(model)  
  
  list(  
    summary_res = summary_model,  
    random_effect_res = rand_eff,
```



```

        fixed_effects_res = fix_eff,
        var_cov_matrix= VarCovMat,
        var_corr_matrix = VarCorrMat,
        Vij_est = Vij
      )
    }

#Diagnostic Plots

diagnostic_plt <- function(df,model){

  residualval <- resid(model)
  fittedval <- fitted(model)

  plot_df <- df %>%
    mutate( resmod = residualval,
            fitmod = fittedval 32 )

  re <- ranef(model)[[1]]
  re_df <- data.frame(random_intercept = as.numeric(re))

  # Residual vs fitted plot

  p1 <- ggplot(plot_df, aes(x = fitmod, y = resmod)) +
    geom_point(alpha = 0.7, color = "grey35") +
    geom_smooth(aes(group = 1), color = "grey20", linewidth = 0.8, se
= FALSE) +
    geom_hline(yintercept = 0, linetype = "dashed", color = "grey50")
  +
    labs( x = "Fitted values",
          y = "Residuals",
          title = "Residuals vs Fitted" ) +
    theme_minimal(base_size = 12) +
  theme(
    panel.grid.minor = element_blank(),
    panel.grid.major = element_line(color = "grey85"),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 11),
    plot.title = element_text(size = 13, face = "bold")
  )

  # QQ Plot for residuals

  p2 <- ggplot(plot_df, aes(sample = resmod)) +
    geom_qq(alpha = 0.7, color = "grey35") +
    geom_qq_line(color = "grey20", linewidth = 0.8) +
    labs(
      x = "Theoretical quantiles",
      y = "Sample quantiles",

```

```

        title = "QQ Plot of Residuals" ) +
    theme_minimal(base_size = 12) +
    theme(
      panel.grid.minor = element_blank(),
      panel.grid.major = element_line(color = "grey85"),
      axis.title = element_text(size = 12),
      axis.text = element_text(size = 11),
      plot.title = element_text(size = 13, face = "bold")
    )

# QQ Plot for random effects

p3 <- ggplot(re_df, aes(sample = random_intercept)) +
  geom_qq(alpha = 0.7, color = "grey35") +
  geom_qq_line(color = "grey20", linewidth = 0.8) +
  labs(
    x = "Theoretical quantiles",
    y = "Sample quantiles",
    title = "QQ Plot of Random Effects" ) +
  theme_minimal(base_size = 12) +
  theme(
    panel.grid.minor = element_blank(),
    panel.grid.major = element_line(color = "grey85"),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 11),
    plot.title = element_text(size = 13, face = "bold")
  )

  list(
    resid_fit = p1,
    fixed_qq_plot = p2,
    re_qq_plot = p3
  )
}

```

Model 1:

```

#create interaction term
aids_sub <- aids_sub %>%
  mutate(Day_cd4 = Day*cd4)

model1_ml <- lme(RNA ~ Day + Day_cd4,
  random = ~1 | patid,
  method = "ML",
  data = aids_sub) #ML used for AIC

model1_reml <- lme(RNA ~ Day + Day_cd4,
  random = ~1 | patid,

```

```

        method = "REML",
        data = aids_sub) #REML used for final estimation

model1_results <- summary_res(model1_reml)
model1_results
diagnostic_plt(aids_sub,model1_reml)

#Blups of beta_i
beta_hat1 <- fixed.effects(model1_reml)
mu_hat1 <- random.effects(model1_reml)

beta_i_model1 <- data.frame(
  patid = rownames(mu_hat1),
  beta1_i = beta_hat1["(Intercept)"] + mu_hat1[, "(Intercept)"],
  beta2_i = beta_hat1["Day"],
  beta3_i = beta_hat1["Day_cd4"] ); beta_i_model1

```

Model 2:

```

model2_ml <- lme(RNA ~ Day + Day_cd4,
  random = ~ Day + Day_cd4 | patid,
  method = "ML",
  data = aids_sub) #ML used for AIC

model2_reml <- lme(RNA ~ Day + Day_cd4,
  random = ~ Day + Day_cd4 | patid,
  method = "REML",
  data = aids_sub) #REML used for final estimation

model2_results <- summary_res(model2_reml)
model2_results
diagnostic_plt(aids_sub,model2_reml)

#Blups of beta_i
beta_hat2 <- fixed.effects(model2_reml)
mu_hat2 <- random.effects(model2_reml)

beta_i_model2 <- data.frame(
  patid = rownames(mu_hat2),
  beta1_i = beta_hat1["(Intercept)"] + mu_hat2[, "(Intercept)"],
  beta2_i = beta_hat1["Day"] + mu_hat2[, "Day"],
  beta3_i = beta_hat1["Day_cd4"] + mu_hat2[, "Day_cd4"]); beta_i_model2

```

Model 3:

```

model3_ml <- nlme(RNA ~ b1*exp(-b2*Day) + b3*exp(-b4*cd4),
  fixed = b1 + b2 + b3 + b4 ~ 1,
  random = b1 ~ 1 | patid,

```

```

        method = "ML",
        data = aids_sub,
        start = c(b1 = 1, b2 = 0.01, b3 = 1, b4 = 0.01)
    )

model3_reml <- nlme(RNA ~ b1*exp(-b2*Day) + b3*exp(-b4*cd4),
    fixed = b1 + b2 + b3 + b4 ~ 1,
    random = b1 ~ 1 | patid,
    method = "REML",
    data = aids_sub,
    start = c(b1 = 1, b2 = 0.01, b3 = 1, b4 = 0.01)
)

model3_results <- summary_res(model3_reml)
model3_results
diagnostic_plt(aids_sub,model3_reml)

#Blups of beta_i
beta_hat3 <- fixed.effects(model3_reml)
mu_hat3 <- random.effects(model3_reml)

beta_i_model3 <- data.frame(
    patid = rownames(mu_hat3),
    beta1_i = beta_hat3["b1"] + mu_hat3[, "b1"],
    beta2_i = beta_hat3["b2"],
    beta3_i = beta_hat3["b3"],
    beta4_i = beta_hat3["b4"]); beta_i_model3

Model 4:

model4_ml <- nlme(RNA ~ b1*exp(-b2*Day) + b3*exp(-b4*Day_cd4),
    fixed = b1 + b2 + b3 + b4 ~ 1,
    random = list(patid = pdDiag(list(b1 ~ 1, b2 ~ 1, b3 ~ 1,
    b4 ~ 1))),
    method = "ML",
    data = aids_sub,
    start = c(b1 = 1, b2 = 0.01, b3 = 1, b4 = 0.01),
    control = nlmeControl(
        maxIter = 200,
        pnlsMaxIter = 50,
        msMaxIter = 200,
        msVerbose = TRUE ) )

model4_reml <- nlme(RNA ~ b1*exp(-b2*Day) + b3*exp(-b4*Day_cd4),
    fixed = b1 + b2 + b3 + b4 ~ 1,
    random = list(patid = pdDiag(list(b1 ~ 1, b2 ~ 1, b3 ~ 1,
    b4 ~ 1))),
    method = "REML",
    data = aids_sub,

```

```

      start = c(b1 = 1, b2 = 0.01, b3 = 1, b4 = 0.01),
      control = nlmeControl(
        maxIter = 200,
        pnlsMaxIter = 50,
        msMaxIter = 200,
        msVerbose = TRUE ) )

model4_results <- summary_res(model4_reml)
model4_results
diagnostic_plt(aids_sub,model4_reml)

#Blups of beta_i
beta_hat4 <- fixed.effects(model4_reml)
mu_hat4 <- random.effects(model4_reml)

beta_i_model4 <- data.frame(
  patid = rownames(mu_hat4),
  beta1_i = beta_hat4["b1"] + mu_hat4[, "b1"],
  beta2_i = beta_hat4["b2"] + mu_hat4[, "b2"],
  beta3_i = beta_hat4["b3"] + mu_hat4[, "b3"],
  beta4_i = beta_hat4["b4"] + mu_hat4[, "b4"]); beta_i_model4

```

Model Comparison:

```

model_aic <- AIC(model1_ml,model2_ml,model3_ml,model4_ml)
model_aic

```